

DUT Software Plane

Operating System

Native NVMe / NIC Drivers

native device contract

Control Plane

DUT-Visible Device Interface

Virtual BDFs, BARs, capabilities

configurable view

Control Virtualization

topology, configuration, routing

compatible control

Selective Passthrough

physical execution after adaptation

device command

Data Plane

DUT Memory

queues and payload buffers

payload

DMA Window / Transport

device DMA

Physical Backend

NVMe SSD or 82580 NIC

control / metadata

payload movement